RAJALAKSHMI ENGINEERING COLLEGE

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M2

Started on	Wednesday, 26 August 2026, 1:27 PM
State	Finished
Completed on	Wednesday, 26 August 2026, 2:38 PM
Time taken	1 hour 10 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);
    int count = 0;

    // Count the initializations (assignments)
    int i = 1;
    count++;  // i = 1
    int s = 1;
    count++;  // s = 1

    while (1) {
        // Count the while condition evaluation
        count++;
        if (!(s <= n)) {
            break;
        }
        // Body
        i++;
        count++;  // i++
        s += i;
        count++;  // s += i
    }

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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